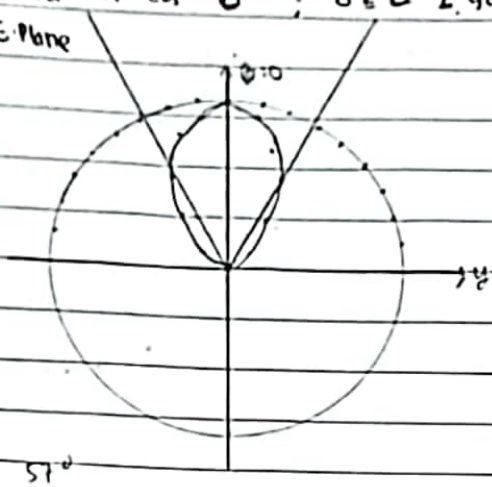
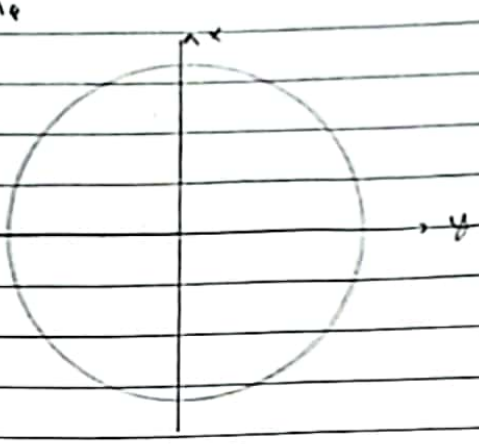
No
Date

1- antenna $U = \cos^4\theta$; $0 \leq \theta \leq 90°$ & $0 \leq \phi \leq 360°$

a. E Plane



H-plane



b- HPBW E-plane & H-plane

HPBW E-plane = 57°

rad = 0,995 rad

HPBW H-plane = 360°

rad = 6,28 rad

d: Directivitas pendekatan

$$D = \frac{41253}{(HPBW_1)(HPBW_2)} = \frac{41253}{(57°)(360°)} = \frac{41253}{20520}$$

$$= 2,01$$

c: Direktivitas eksak

$$D = \frac{4\pi U_{max}}{P_{rad}} \rightarrow P_{rad} = \int_0^{\pi} \int_0^{57/2} U \sin\theta \, d\theta \, d\phi$$

$$= \frac{4\pi \cdot 1}{0,12} = \int_0^{\pi} d\phi \int_0^{\pi} \cos^4\theta \sin\theta \, d\theta$$

$$= 20,83. = -\frac{1}{5}\cos^5\theta \Big|_0^{\pi/2}$$

$$= \frac{1}{5}$$

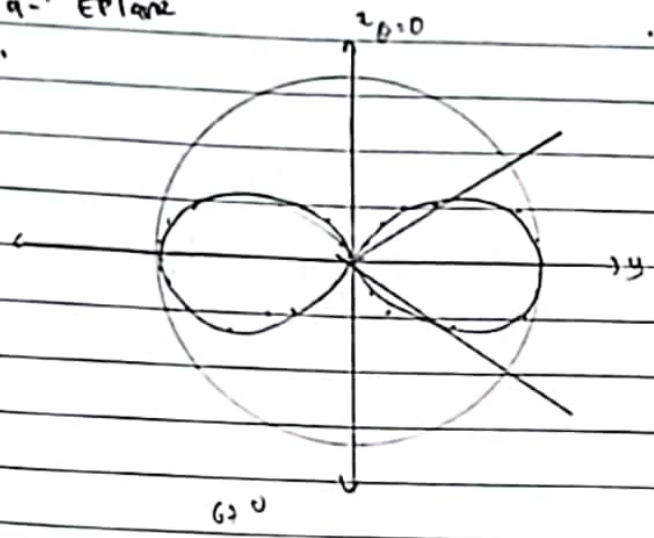
e. Perbandingan direktivity antara eksak dan pendekatan yang cukup jauh

f. Pola radiasi : Directional

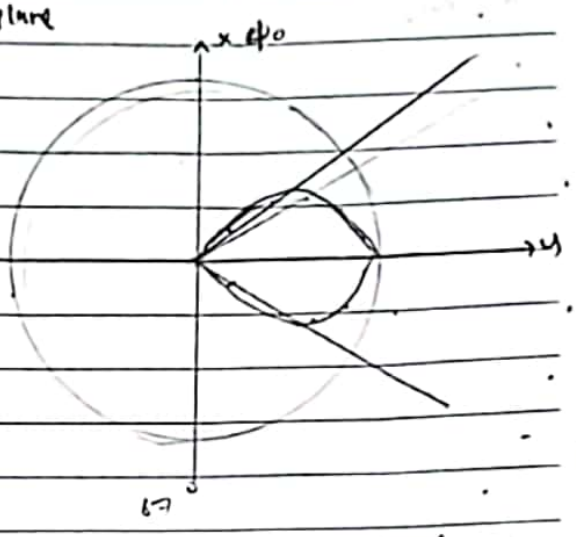
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3. antenna dng intensitas $U = \sin^2\theta \sin^3\phi$; $0 \le \phi \le 2\pi$; $0 \le \theta \le \pi$

a- E-plane



H-plane



b- HPBW E-plane & H-plane

HPBW E-plane = 66°

rad = 1,152

HPBW H-plane = 69°

rad = 1,2

d- Direktivitas Perdekatan

$$D = \frac{41253}{(HPBW_1)(HPBW_2)} = \frac{41253}{(66^\circ)(69^\circ)} = \frac{41253}{4554} = 9,05$$

c- Direktivitas Eksak

$$D = \frac{4\pi U_{max}}{P_{rad}} \rightarrow P_{rad} = \int_0^{2\pi} \sin^3\phi \, d\phi \int_0^{\pi} \sin^3\theta \, d\theta$$

$$= \frac{4\pi \cdot 1}{-1,7}$$

$$= -7,39$$

$$= \cos\phi - \frac{1}{3}\cos^3\phi \Big|_{\pi}^{2\pi} \cos\theta - \frac{1}{3}\cos^3\theta \Big|_0^{\pi}$$

$$= \frac{2}{3} - \left(-\frac{2}{3}\right) \cdot \left(-\frac{3}{3}\right) = \frac{2}{3}$$

$$= \frac{4}{3} \cdot \frac{-4}{3} = \frac{-16}{9} = -1,7$$

e- Perbandingannya antara Eksak dan pendekatan cukup berbeda

f. pola radiasi : direktional

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3. Dik : P_t = Pwatt

G_t = 70 → 18,45 dB

e = 85 %

F = 2,4 GHz = 2400 MHz

P_r = 10^{-6} mW → −60 dBm

G_r = 17 dB

e = 90 %

Dit : daya pemancar (P_t) ?

Dijawab :

$P_r = P_t$(dBm) + G_t(dB) + G_r(dB) − 20 log d(km) − 20 log F(MHz) − 32,45

−60 = P_t + 18,45 + 17 − 20 log 3 − 20 log 2400 − 32,45

−60 = P_t + 35,45 − 9,54 − 67,6 − 32,45

−60 = P_t − 74,14

P_t = 14,14 dBm

= 25,94 mW

4. Dik : h = 3mm

ε_r = 4,4

F = 5,5 GHz → $5,5 \cdot 10^9$ Hz

t = 0,035 mm

c = $3 \cdot 10^8$ m/s = $3 \cdot 10^{11}$ mm/s

Jawab :

① $$W = \frac{c}{2f}\sqrt{\frac{2}{\varepsilon_r + 1}}$$

$$= \frac{3 \cdot 10^{11}}{2 \cdot 5,5 \cdot 10^9}\sqrt{\frac{2}{4,4+1}}$$

$$= 27,27 \cdot 0,608$$

$$= 16,58 \text{ mm}$$

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$$\varepsilon_{reff} = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2}\left(\left[1 + \frac{12.h}{w}\right]^{-1/2}\right)$$

$$= \frac{4,4+1}{2} + \frac{4,4-1}{2}\left(\left[1 + \frac{12.3}{16,50}\right]^{-1/2}\right)$$

$$= 2,7 + 1,7\ (0,56)$$

$$= 2,7 + 0,952$$

$$= 3,652$$

$$\Delta L = 0,412.h\ \frac{(\varepsilon_{reff} + 0,3)(w/h + 0,264)}{(\varepsilon_{reff} - 0,258)(w/h + 0,8)}$$

$$= 0,412.3\ \frac{(3,652 + 0,3)(16,50/3 + 0,264)}{(3,652 - 0,258)(16,50/3 + 0,8)}$$

$$= 1,236\ \frac{(3,952)(5,79)}{(3,394)(6,32)}$$

$$= \frac{20,28}{21,45}$$

$$= 1,31 \text{ mm}$$

$$L = \frac{c}{2f\sqrt{\varepsilon_{reff}}} - 2\Delta L$$

$$= \frac{3.10^{11}}{2.5,5.10^{9}\sqrt{3,652}} - 2.1,31$$

$$= 14,27 - 2,62$$

$$= 11,65 \text{ mm}$$

(b) $Z_0 = \frac{c}{f}$ $\quad$ Dik : $Z_0 = 100\ \Omega$

$$= \frac{3.10^{11}}{5,5.10^{9}}$$

$$= 54,54 \text{ mm} \longrightarrow w << Z_0 \rightarrow G_1 = \frac{1}{90}\left(\frac{w}{Z_0}\right)^2$$

$$G_1 = \frac{1}{90}\left(\frac{w}{Z_0}\right)^2$$

$$= \frac{1}{90}\left(\frac{16,50}{54,54}\right)^2$$

$$= 0,001\ \mho$$

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$Z_A = \frac{1}{2 \cdot G_1}$

$= \frac{1}{2 \cdot 0,001}$

$= 500 \Omega$

$Z_B = Z_A \cos^2 \left(\frac{\pi}{L} y_0\right)$

$100 = 500 \cos^2 \left(\frac{3,14}{11,65} y_0\right)$

$0,2 = \cos^2 (0,269\, y_0)$

$0,44 = \cos (0,269\, y_0)$

$1,715 = 0,269\, y_0$

$y_0 = 4,14$

$l_g = 6h + L = 6,3 + 11,65 = 29,65$ mm

$W_g = 6h + w = 6,3 + 16,50 = 34,58$ mm

③

